

Microprocessor & Computer Architecture

BCA — Semester II — 2020 — Answer Sheet

Subject Code: CACS155

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. Define CPU. Differentiate between Microprocessor and Microcontroller with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define instruction cycle. Explain the opcode fetch machine cycle for MOV A, B with timing diagram. (Opcode: MOV A, B = 78h)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Write an ALP using 8085 to check number stored in memory location 8080h is either even or odd.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is cache memory? Explain the elements of cache design.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain the organization of Microprogrammed Control Unit.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is pipeline? Explain the four segment instruction pipeline.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write short notes on (any two): a) Accumulator b) 8085 Interrupts c) Structure of Hard Disk

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define instruction set. Classify the instructions available in 8085 with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define the addressing mode. Explain the various instruction addressing modes with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define micro-program? Describe symbolic micro-program for instruction FETCH routine. Explain the organization of micro-program sequencer for control memory with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.